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FDP SESSION 2026

CLOUD × AI

Cloud Computing for AI

Faculty Development Program Session — bridging elastic cloud infrastructure with modern AI workloads.

PRESENTED BY

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DATE

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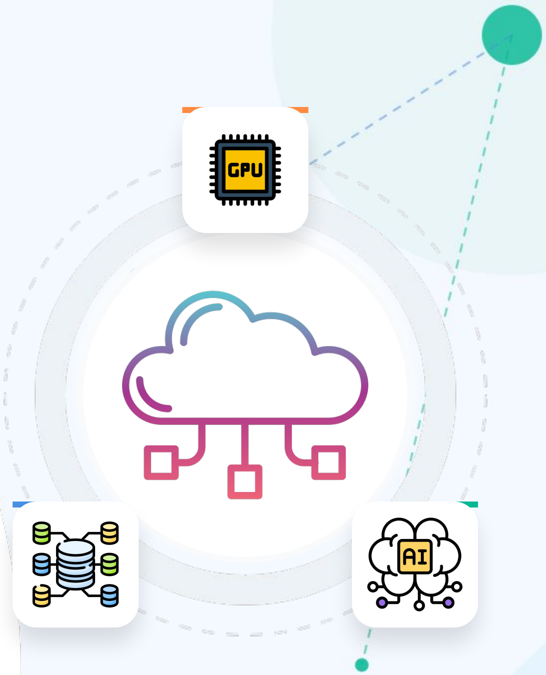
Faculty Development Program



AGENDA

Session Agenda

A comprehensive roadmap bridging cloud fundamentals, elastic infrastructure, and hands-on MLOps deployments.

Phase 1 · Foundations

01 · Cloud Fundamentals
NIST definition, characteristics, evolution.

02 · Service & Deployment
IaaS, PaaS, SaaS · Public, Private, Hybrid.

03 · Cloud Providers for AI
AWS, GCP, Azure, and IBM compared.

Phase 2 · Infrastructure

04 · GPU / TPU Computing
Specialized high-performance hardware on-demand.

05 · Cloud AI/ML Platforms
SageMaker, Vertex AI, and Azure ML.

06 · Storage & Data Services
S3, distributed data lakes, and feature stores.

Phase 3 · Operations

07 · Training & Deployment
Distributed training, secure containers, robust APIs.

08 · MLOps in the Cloud
CI/CD pipelines, versioning, drift monitoring.

09 · Case Study & Hands-On
AWS AI Search and Lightning.ai free tier platform.

What is Cloud Computing?

NIST DEFINITION

A model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources — networks, servers, storage, applications, and services — that can be rapidly provisioned and released with minimal management effort or service-provider interaction.

Delivered over the internet

Resources are accessed remotely, not owned physically by the user.

Pay only for what you use

No large upfront hardware investment; OpEx instead of CapEx.

Elastic & self-service

Provision compute, storage, or networking in minutes, scale up or down on demand.

INTUITION

Electricity Grid

Plug in · pay for usage · no power plant to own

=

Cloud Computing

Rent compute · pay per hour · no data center to own

"Cloud is the electricity grid for computing."

THE WHY

Why AI Needs the Cloud

Modern AI workloads are compute-hungry, data-heavy, and bursty — exactly the shape cloud was built to serve.

01

Massive Elastic Compute

Training large models needs thousands of GPU-hours. The cloud lets you spin up 100 GPUs for a week, then shut them down — instead of buying hardware that sits idle 90% of the time.

GPUs on-demand · released when done

02

Cost-Efficient by Design

On-prem GPU clusters are cost-prohibitive for most institutions. Pay-as-you-go cloud pricing converts large capital expenditure (CapEx) into predictable operational expenditure (OpEx).

CapEx → OpEx

03

Specialized Hardware Access

Cloud providers refresh GPU/TPU generations faster than most organizations can buy them — giving teams instant access to cutting-edge silicon like NVIDIA H100, Google TPU v5, AWS Trainium.

Latest silicon · zero procurement

EVOLUTION

From Data Centers to Cloud AI

Cloud didn't start as an AI tool — it evolved to meet AI's compute demands as deep learning took off.

On-Prem

- Owned physical servers
- Fixed capacity & high CapEx
- Manual provisioning

Hardware bought & racked

Virtualization

- Multiple VMs per host
- Higher hardware utilization
- Snapshot & migrate jobs

Decouple OS from hardware

Cloud

Onward

- AWS launches EC2 & S3
- Rented by the hour
- Self-service & elastic

Compute as a utility

AI Cloud

Onward

- AlexNet sparks AI boom
- GPU/TPU cloud instances
- Custom silicon chips

Cloud built for AI

Key Milestone: 2012 deep-learning breakthrough triggered cloud providers to pivot rapidly toward AI-optimized hardware.

Core Characteristics

The five traits that distinguish cloud from traditional hosting — each with a direct AI angle.

01

On-Demand Self-Service

A data scientist can provision a GPU instance in minutes — no IT ticket, no human approval step required.

02

Broad Network Access

Models, datasets, and APIs are reachable from anywhere over standard internet protocols — laptops, phones, edge devices.

03

Resource Pooling

Multiple customers share the same physical infrastructure (multi-tenancy), with resources dynamically assigned based on demand.

04

Rapid Elasticity

Auto-scale training clusters from 1 to 64 GPUs in minutes during a job, then scale back to zero when training ends. Capabilities appear unlimited to the user.

AI ANGLE · TRAINING CLUSTERS

05

Measured Service

Billing is metered precisely — GPU-hours, storage-GB, API calls. You see exactly what you consumed and pay only for that.

AI ANGLE · METERED GPU-HOURS

SERVICE MODELS

IaaS · PaaS · SaaS

As you move IaaS → PaaS → SaaS, you gain convenience but lose control. Each has a natural AI use case.

MOST CONTROL

IaaS — Infrastructure as a Service

YOU MANAGE OS + SOFTWARE

WHAT YOU MANAGE

Rent raw compute, storage, networking. The provider manages only the physical hardware — you handle OS, runtime, frameworks, and your model code.

AI EXAMPLE

Renting a raw GPU VM (e.g., AWS EC2 P4 instance) and installing your own PyTorch / TensorFlow stack from scratch.

EC2 GPU VM | Bare Metal

BALANCED

PaaS — Platform as a Service

PROVIDER MANAGES INFRA

WHAT YOU MANAGE

Bring your code and data — the platform handles provisioning, scaling, orchestration, and runtime. Infrastructure is fully abstracted away.

AI EXAMPLE

Amazon SageMaker or Google Vertex AI: managed notebooks, distributed training jobs, and one-click model endpoints.

SageMaker | Vertex AI

MOST CONVENIENT

SaaS — Software as a Service

PROVIDER MANAGES EVERYTHING

WHAT YOU MANAGE

Nothing — just use the ready-made application through a browser or API. No infrastructure or platform to manage at all.

AI EXAMPLE

A ready-made AI assistant like ChatGPT or a hosted document-intelligence API — you only call it.

ChatGPT | Hosted AI APIs

Teaching point: more convenience → less control — pick the layer that matches your team's skills and use case.

Public · Private · Hybrid · Multi-Cloud

Four ways to deploy cloud infrastructure — each with a different trade-off between cost, control, and compliance for AI workloads.

LOWEST COST

Public Cloud

Shared infrastructure offered by AWS / Azure / GCP. Multi-tenant, billed by usage, no upfront commitment.

PRO

Cheapest · fastest to start.

CON

Least control · shared tenancy.

FULL CONTROL

Private Cloud

Dedicated infrastructure for a single organization — on-prem or hosted. Required for highly sensitive AI data (e.g., healthcare, defense).

PRO

Max control · compliance-friendly.

CON

Expensive · slower to scale.

BALANCED

Hybrid Cloud

Mix of public + private — train models in public cloud, keep sensitive data on private infrastructure. Common in regulated industries.

PRO

Best of both worlds.

CON

Network & data-flow complexity.

NO LOCK-IN

Multi-Cloud

Use more than one public cloud — e.g., AWS for storage, GCP for a specific AI API. Reduces vendor lock-in but increases management overhead.

PRO

Avoids vendor lock-in.

CON

Higher operational complexity.

Key Cloud Providers for AI

No single “best” provider — the right choice depends on existing infrastructure, budget, compliance needs, and team familiarity.

BROADEST CATALOG

Amazon Web Services

INDUSTRY LEADER · DEEPEST CATALOG

Broadest service catalog with SageMaker for end-to-end ML, plus the dominant S3 / EC2 ecosystem. Widely adopted across industry.

- SageMaker end-to-end ML
- S3 · EC2 · Lambda ecosystem
- Custom Trainium / Inferentia

RESEARCH-GRADE ML

Google Cloud Platform

TENSORFLOW NATIVE · TPUS

Vertex AI unifies AutoML and custom training. Exclusive access to TPUs — custom AI chips for tensor ops — and strong research-grade ML tooling.

- Vertex AI unified platform
- Exclusive TPU pods
- BigQuery + ML integration

ENTERPRISE-READY

Microsoft Azure

ENTERPRISE INTEGRATION · OPENAI

Azure ML Studio pairs a visual designer with code-first notebooks. Deep integration with enterprise tools and the exclusive OpenAI partnership.

- Azure ML Studio · designer
- OpenAI hosted models
- Azure DevOps MLOps

HYBRID & WATSON

IBM Cloud

WATSON AI · HYBRID FOCUS

Watson AI services cover NLP, speech, and vision. Strong enterprise and hybrid-cloud story — appealing to highly regulated industries.

- Watson AI services
- Red Hat OpenShift hybrid
- Quantum-safe security

GPU / TPU Computing in the Cloud

WHY SPECIALIZED HARDWARE MATTERS

CPU Sequential processing

A handful of powerful cores — great for general tasks, but slow for the matrix multiplications at the heart of neural networks.

GPU Thousands of parallel cores

Massively parallel — ideal for the matrix multiplies inside deep-learning training. Originally built for graphics, now the workhorse of AI.

TPU Custom tensor silicon

Tensor Processing Units — Google-built chips purpose-designed for tensor operations in deep learning. Highest throughput for compatible workloads.

CLOUD ADVANTAGE

On-Demand GPU Clusters

Rent exactly the number of GPUs you need, only for the duration of training — no idle hardware.

Spot / Preemptible Instances

Discounted, interruptible GPU access for non-urgent jobs — up to 60–90% cheaper than on-demand.

Pay for compute only when training is running.

MANAGED AI PLATFORMS

Cloud AI/ML Platforms

All three abstract away infrastructure so teams can focus on data and models rather than server provisioning.

Amazon SageMaker

PROVIDER · AWS

WHAT IT IS

End-to-end ML platform covering the full lifecycle — data labeling, distributed training, hyperparameter tuning, and production monitoring.

KEY FEATURES

- Built-in algorithms & custom frameworks
- Managed notebooks & training jobs
- Automatic model tuning
- One-click endpoint deployment

End-to-end · zero infra

Google Vertex AI

PROVIDER · GOOGLE CLOUD

WHAT IT IS

Unified platform combining AutoML (no-code) with custom training. Native BigQuery integration for data-heavy pipelines and exclusive TPU access.

KEY FEATURES

- AutoML for tabular, image, text & video
- Custom training with PyTorch, TF, JAX
- BigQuery & Feature Store integration
- Model Registry & Endpoints

AutoML + TPU native

Azure ML Studio

PROVIDER · MICROSOFT AZURE

WHAT IT IS

Drag-and-drop visual pipeline designer alongside code-based notebooks. Deep MLOps integration with Azure DevOps and GitHub Actions.

KEY FEATURES

- Visual designer + code notebooks
- Automated ML & hyperparameter tuning
- Azure DevOps native CI/CD
- Responsible AI dashboard built-in

Visual + code MLOps

DATA LAYER

Storage & Data Services for AI

The data layer feeds every ML pipeline — from raw ingestion through feature engineering to model artifacts.

UNSTRUCTURED

Object Storage (S3/GCS)

Stores unstructured data — images, videos, audio, raw text — at massive scale and low cost per GB.

EXAMPLES

Amazon S3
Google Cloud
Azure Blob

RAW · CENTRAL

Data Lakes

Centralized repositories holding raw data in its native format until needed. Ideal for AI workloads.

EXAMPLES

Lake Formation
Delta Lake
Iceberg

STRUCTURED

Managed Databases

Structured / relational data feeding feature-engineering pipelines. Cloud providers handle scaling.

EXAMPLES

Amazon RDS
Cloud SQL
Azure SQL

CONSISTENCY

Feature Stores

Centralized repository for engineered features — ensuring the same feature definitions are used globally.

EXAMPLES

SageMaker FS
Feast
Vertex FS

Cloud-based Model Training

Cloud training turns weeks of single-machine training into hours — through parallelism, elasticity, and discounted compute.

Distributed Training

Split a training job across multiple GPUs or nodes to slash wall-clock training time. Two common strategies: data parallelism (same model, different data shards) and model parallelism (model itself split across devices).

Scales linearly with GPU count

Auto-Scaling Clusters

Cloud platforms automatically add or remove compute nodes based on training-job demand. A 64-GPU job spins up only for the run, then drops to zero — you never pay for idle capacity between jobs.

Scale up on demand · scale to zero

Spot Instances

Use discounted, interruptible cloud compute for non-time-critical training jobs. Spot pricing can cut training costs by 60–90% vs on-demand — at the cost of possible mid-job interruption (handled via checkpointing).

Up to 90% cost reduction

Cloud-based Model Deployment

A trained model is only useful when it's serving predictions — the cloud provides the serving fabric to make that reliable and scalable.

Model Serving (API Endpoints)

Once trained, a model is exposed so applications can send it data and get predictions back — typically via a REST or WebSocket API endpoint managed by a serving framework.

REST / WEBSOCKET ENDPOINTS

Containers & Kubernetes

Models are packaged in Docker containers for consistent deployment across dev, test, and production. Kubernetes orchestrates them — scaling inference workloads up and down based on request volume.

DOCKER + K8S · PORTABLE

Edge–Cloud Hybrid Inference

Lightweight inference runs on edge devices (phones, cameras, IoT) for ultra-low latency, while heavier or batch processing happens in the cloud. Balances speed (edge) with power (cloud).

LATENCY ON EDGE · POWER IN CLOUD

Pattern:containerized model behind an API, auto-scaled by request load — the same shape used in our Slide 16 case study.

MLOps in the Cloud

Applying DevOps principles (CI/CD, automation, monitoring) to the ML lifecycle — without it, models become "science projects" that never reliably reach production.

CI/CD for ML

AUTOMATION

Automated testing and deployment of new model versions.
Every code change triggers tests, training runs, and controlled rollout.

GitHub Actions | GitLab CI

Pipeline Automation

PIPELINE

Automating the full sequence from data preprocessing to deployment.
Retraining becomes fully reproducible at the push of a button.

Kubeflow | Airflow

Model Versioning

VERSIONING

Tracking production models with full lineage back to training data.
Enables instant rollback if a newly deployed version underperforms.

MLflow | Model Registry

Monitoring for Drift

MONITORING

Detecting when model performance degrades due to real-world data drift.
Automatically triggering alerts and orchestrating retraining cycles.

Evidently | SageMaker Monitor

AI Search Project — End-to-End on AWS

A real video-search system: client uploads a video, sends a natural-language query, and gets the top 5 matching moments back in real time.

1 Video Upload

Client sends video via REST API (/upload_ai_search_video).

2 Amazon S3 Storage

Video stored in S3 bucket (ai-search-video); access URL returned.

3 WebSocket Connection

Client opens /ws/ai_search/{id}, sends search_text + video URL.

4 Frame Extraction (OpenCV)

FPS-based smart batching — processes ~1 batch/sec, not every frame.

5 AI Inference Model

Batch of frames + search text → matches with confidence scores.

6 Top 5 Results Returned

Timestamps + confidence scores pushed back via WebSocket; earliest first.

Why AWS?

S3 Storage: Scalable, durable video storage at ~$0.023/GB.

Co-location: Region-based deployment keeps storage & compute co-located for low latency.

EC2/GPU Compute: Runs the FastAPI backend + AI inference with no owned servers.

The Pattern *Object storage + compute + real-time connection + AI inference.*

HANDS-ON PRACTICE

Free Platform — Lightning.ai

You just saw a production AWS example. Want to build and deploy your own AI models — for free?

From watching AWS → to deploying your own model.

No cloud billing. No local setup. Just open a browser and start building.

Free GPU Compute

Free GPU-backed compute for training and experimentation — no need to provision cloud infrastructure yourself or worry about surprise bills.

FREE GPU HOURS

AI Studio Environments

Pre-built, cloud-hosted dev environments that launch instantly in your browser — no local Python, CUDA, or driver setup required.

ZERO LOCAL SETUP

Ready-Made Templates

Templates for the most common AI workflows — fine-tuning a model, building an inference API, deploying a model-serving endpoint.

1-CLICK TEMPLATES

Built for Learning

Lowers the barrier for students & faculty to go from "learning about cloud AI" to "actually deploying a model" — in a single session.

CLASSROOM-READY

Let's explore the platform →

WRAP-UP

Summary, Future Trends & Q&A

KEY TAKEAWAYS

1 **Cloud = AI's compute backbone**
Elastic, on-demand GPU/TPU access is what makes modern AI training affordable.

2 **Models = trade-offs**
IaaS/PaaS/SaaS and public/private/hybrid each balance control, cost, and convenience.

3 **A repeatable pattern**
Storage + compute + data pipeline + inference — the shape of every production cloud-AI system.

4 **Free practice is real**
Lightning.ai lets you start deploying models today — no budget required.

WHAT'S NEXT

Edge–Cloud Convergence
More inference moves to edge devices (phones, cameras, IoT) for low latency; the cloud handles training and heavier processing.

Sustainable / Green AI
Providers optimize data centers for energy efficiency as AI's power demands grow — carbon-aware training is becoming standard.

Serverless AI Inference
Pay-per-request model serving with zero server management — invoke an endpoint, pay only for the milliseconds of compute used.

Questions & Discussion

Open floor — what would you like to dig deeper into?

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Thank You.

Questions, ideas, or want to build something together — let's talk.

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Cloud + AI = the new default. Let's build.